

Mathematical methods for a European bank Credit-Risk Analysis

Companion note to the GitHub project

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This note explains the mathematical ideas used in a small quantitative credit-risk project based on public European Banking Authority (EBA) data, with Barclays Bank Ireland plc used as the principal case study. It is written for readers who may be new to finance. Each section first explains the financial question in ordinary language and then introduces the corresponding calculation. The analysis covers exposure composition, defaulted-exposure ratios, concentration measures, changes through time, peer comparison, and a simple hypothetical stress-sensitivity exercise. These calculations are descriptive and educational: they do not reproduce an internal bank credit model, an expected-credit-loss model, or an official regulatory stress test.

I. WHAT THIS PROJECT IS TRYING TO MEASURE

A bank lends money and enters into other transactions with governments, companies, financial institutions, households, and other counterparties. These activities create *credit exposure*: an amount that may be at risk if a borrower or counterparty cannot meet its obligations.

The EBA dataset groups these exposures into regulatory exposure classes. In this project, the calculations are designed to answer a sequence of simple questions:

1. How large is the selected exposure portfolio?
2. Which exposure classes account for most of it?
3. How much of the selected exposure is reported as defaulted?
4. Is the exposure spread across many classes or concentrated in only a few?
5. How have these quantities changed across reporting dates?
6. How does Barclays Bank Ireland compare descriptively with selected peers?
7. What happens mathematically if the amount of defaulted exposure is increased under hypothetical stress assumptions?

The word “risk” should therefore be interpreted carefully throughout this note. The project studies selected public regulatory exposure data. It does not calculate the probability that an individual borrower will default and it does not estimate the bank’s future losses.

II. NOTATION AND BASIC FINANCIAL QUANTITIES

Before performing calculations, it is useful to define the symbols used throughout the analysis.

For reporting period t , let

$$E_{i,t} = \text{exposure value of class } i, \tag{1}$$

$$E_{\text{tot},t} = \sum_{i=1}^n E_{i,t}, \tag{2}$$

$$E_{\text{def},t} = \text{exposure classified as defaulted}. \tag{3}$$

Here, i identifies an exposure class and t identifies a reporting date. The symbol n is the number of exposure classes included in the selected dataset slice.

A. What is an exposure class?

An exposure class is a regulatory grouping of exposures with broadly similar counterparty or transaction characteristics. Examples in the selected EBA data include central governments or central banks, corporates, institutions, public-sector entities, and defaulted exposures.

Suppose, for illustration, that a selected portfolio contains exposure values of 60, 30, and 10 units in three classes. The total exposure is simply

$$E_{\text{tot}} = 60 + 30 + 10 = 100. \quad (4)$$

The first class therefore represents 60% of the selected total, the second 30%, and the third 10%. Much of this project generalizes this simple idea to the actual EBA data.

III. EXPOSURE SHARES: UNDERSTANDING THE COMPOSITION OF THE PORTFOLIO

A. Financial question

The absolute size of an exposure class is useful, but it does not immediately tell us how important that class is relative to the whole selected portfolio. The first calculation therefore asks:

What fraction of the selected total exposure belongs to each exposure class?

This converts absolute exposure values into comparable portfolio weights.

B. Calculation

The normalized share of exposure class i is

$$s_{i,t} = \frac{E_{i,t}}{E_{\text{tot},t}}. \quad (5)$$

If an exposure class contains 20,000 units and the selected total exposure is 50,000 units, then

$$s_i = \frac{20000}{50000} = 0.40. \quad (6)$$

The class therefore represents 40% of the selected exposure.

To express the same quantity as a percentage,

$$S_{i,t} = 100s_{i,t}. \quad (7)$$

The normalized shares must add to one:

$$\sum_{i=1}^n s_{i,t} = 1, \quad (8)$$

or, equivalently, the percentage shares must add to 100%.

C. Why this calculation is useful

Normalization allows portfolios of different absolute sizes to be compared on a common scale. A bank may have a very large total exposure, but the analytical question here is often about *composition*: for example, whether 20%, 50%, or 80% of the selected exposure is concentrated in one regulatory class.

In the June 2025 Barclays Bank Ireland slice used in this project, the largest selected exposure classes include central governments or central banks, corporates, and institutions. Figure 1 shows the corresponding exposure values. The horizontal bars are ordered by size, so a novice reader can immediately see that a small number of regulatory classes account for most of the selected exposure. The figure should be interpreted as a description of *where the exposure sits*; it does not say that the largest class is automatically the riskiest class.

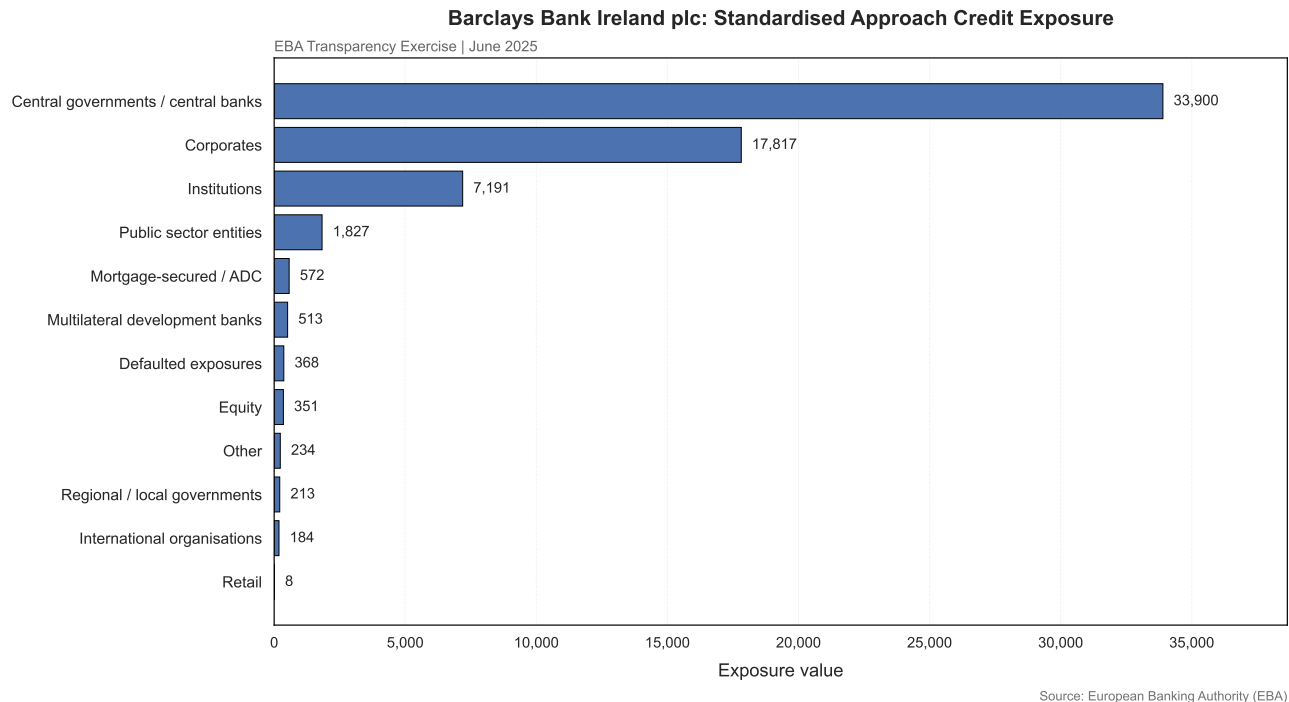


FIG. 1. **Standardised Approach exposure by regulatory exposure class for Barclays Bank Ireland plc in June 2025.** The length of each horizontal bar is the reported exposure value in the selected EBA data slice. Central governments or central banks form the largest class, followed by corporates and institutions. Smaller categories, including defaulted exposures, contribute much less to the total selected exposure. This plot is a composition chart: it shows the relative scale of exposure classes and should not be interpreted as a ranking of their credit quality, probability of default, or expected loss.

IV. DEFAULTED-EXPOSURE SHARE

A. What does “defaulted exposure” mean in this project?

The dataset contains exposure reported in a defaulted category. For this project, the important descriptive question is not only the absolute amount of such exposure, but also its size relative to the selected total exposure.

A value such as 300 has little meaning by itself. If total exposure is 1,000, it represents 30%. If total exposure is 100,000, it represents only 0.3%. The denominator is therefore essential.

B. Calculation

The observed defaulted-exposure share is

$$D_t = \frac{E_{\text{def},t}}{E_{\text{tot},t}} \times 100. \quad (9)$$

For June 2025, the selected Barclays Bank Ireland values are

$$E_{\text{tot}} = 63176.872910, \quad (10)$$

$$E_{\text{def}} = 367.874881. \quad (11)$$

Therefore,

$$D = \frac{367.874881}{63176.872910} \times 100 \approx 0.582294\%. \quad (12)$$

In words, approximately 0.582% of the selected exposure in this particular EBA slice is represented by the defaulted-exposure amount used in the analysis.

C. What this number does and does not mean

This ratio is useful as a descriptive indicator for comparing reporting periods or institutions within the same analytical framework. However,

$$D_t \neq PD. \quad (13)$$

Here PD means *probability of default*. A probability of default is a forward-looking credit-risk parameter that estimates the likelihood that a borrower will default over a specified horizon. The quantity D_t calculated here is instead a ratio constructed from reported aggregate exposure values. It should not be interpreted as the probability that 0.582% of borrowers will default.

V. EXPOSURE-CLASS CONCENTRATION

Exposure shares tell us the size of individual classes. Concentration measures summarize the entire distribution into one or a few numbers.

A. Why concentration matters

Consider two simplified portfolios:

- Portfolio A has four exposure classes, each with a 25% share.
- Portfolio B has one class with 85% and three classes with 5% each.

Both portfolios contain four classes, but they are structurally very different. Portfolio A is evenly distributed, whereas Portfolio B is dominated by one class.

The project therefore measures *exposure-class concentration*. This does not directly measure credit quality. It measures how unevenly the selected exposure is distributed across regulatory classes.

B. Herfindahl–Hirschman Index

The main concentration statistic used is the Herfindahl–Hirschman Index, abbreviated HHI:

$$HHI_t = \sum_{i=1}^n s_{i,t}^2. \quad (14)$$

The calculation has three steps:

1. calculate each exposure-class share s_i ;
2. square every share;
3. add the squared shares.

Squaring is important because it gives disproportionately more weight to large shares. For four equally weighted classes,

$$HHI = 4 \left(\frac{1}{4} \right)^2 = 0.25. \quad (15)$$

Now consider a much more concentrated example with shares 0.85, 0.05, 0.05, and 0.05:

$$HHI = 0.85^2 + 0.05^2 + 0.05^2 + 0.05^2 \quad (16)$$

$$= 0.73. \quad (17)$$

The second HHI is much larger because one class dominates the distribution. At the theoretical extreme, if one class contains all selected exposure,

$$HHI = 1. \quad (18)$$

Thus, within the same definition of classes, a higher HHI means a more concentrated distribution and a lower HHI means a more dispersed distribution.

C. HHI in the project

For the selected Barclays Bank Ireland exposure-class data, the calculated HHI values are approximately

Reporting period	HHI
September 2024	0.337
December 2024	0.336
March 2025	0.396
June 2025	0.381

The increase from about 0.336 in December 2024 to about 0.396 in March 2025 means that the selected exposure distribution became more concentrated across exposure classes. The decrease to about 0.381 in June 2025 indicates some reduction from the March peak, although concentration remained above the September and December 2024 levels. The same movement is visualized in Fig. 2; the sharp rise between December and March is much easier to recognize graphically than from a table alone.

This statement concerns *exposure composition*; it does not by itself imply that the bank became safer or riskier.

D. Top- k concentration

HHI uses every exposure class. A simpler complementary question is:

How much of the selected portfolio is contained in its largest few exposure classes?

Order the shares from largest to smallest,

$$s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(n)}. \quad (19)$$

The top- k concentration ratio is

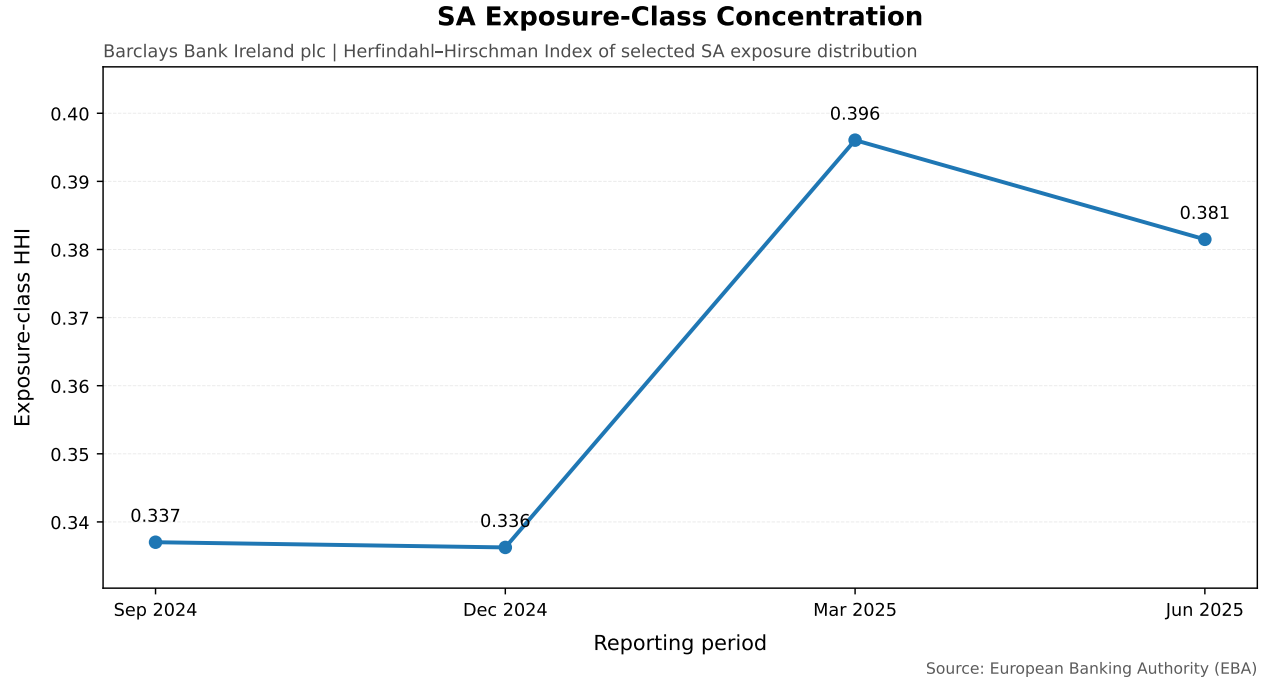


FIG. 2. **Exposure-class concentration through time, measured with the Herfindahl–Hirschman Index (HHI).** HHI is calculated by squaring each exposure-class share and summing the squared shares. A larger value therefore means that a greater fraction of the selected exposure is concentrated in a smaller number of classes. The HHI is almost unchanged between September and December 2024, rises markedly in March 2025, and then moderates in June 2025. The plot measures concentration of regulatory exposure classes; it is not a direct measure of default risk or loss severity.

$$C_k = 100 \sum_{j=1}^k s_{(j)}. \quad (20)$$

For the three largest exposure classes,

$$C_3 = 100 [s_{(1)} + s_{(2)} + s_{(3)}]. \quad (21)$$

If the three largest classes have shares of 50%, 25%, and 15%, then

$$C_3 = 50 + 25 + 15 = 90\%. \quad (22)$$

This means that only three classes account for 90% of the selected exposure. The largest-class share is simply

$$C_1 = 100s_{(1)}. \quad (23)$$

In June 2025, the project finds a top-three share of approximately 93.24% and a largest-class share of approximately 53.66% for the selected exposure slice. These values help explain why the HHI is relatively high: a large fraction of the exposure is located in only a few classes.

E. Difference between HHI and top-three share

The two measures answer related but different questions. C_3 asks how much exposure is contained specifically in the three largest classes. HHI considers every class and is also sensitive to how unevenly the large shares are distributed. For that reason, reporting both measures gives a clearer picture than relying on either one alone.

VI. TIME-SERIES COMPARISON

A. Why compare reporting periods?

A single reporting date provides only a snapshot. Credit-risk analysis often becomes more informative when the same measure is followed through time.

The project contains four reporting periods:

September 2024, December 2024, March 2025, June 2025.

For each date, the project calculates several indicators. They can be summarized mathematically as

$$\mathbf{x}_t = \begin{pmatrix} E_{\text{tot},t} \\ E_{\text{def},t} \\ D_t \\ HHI_t \\ C_{3,t} \end{pmatrix}. \quad (24)$$

This notation simply means that each reporting date has a collection of numbers describing total exposure, defaulted exposure, defaulted share, HHI, and top-three concentration.

B. Defaulted-exposure trend

The observed values are

Period	Total exposure	Defaulted exposure	Defaulted share
Sep. 2024	67799.24	424.60	0.626%
Dec. 2024	68330.28	334.18	0.489%
Mar. 2025	63167.06	281.07	0.445%
Jun. 2025	63176.87	367.87	0.582%

The defaulted-exposure share declined from September 2024 to March 2025 and then increased in June 2025. Figures 3 and 4 show the same development in two complementary ways. Figure 3 plots the absolute amount classified as defaulted, whereas Fig. 4 divides that amount by the selected total exposure. Looking at both is important: the first answers “how much defaulted exposure is reported?”, while the second answers “how large is that amount relative to the selected exposure base?”

This is a descriptive observation from four reporting dates; it should not be interpreted as a statistical forecast of a longer-term trend.

C. Percentage points versus percent change

If the defaulted-exposure share changes from D_{t_1} to D_{t_2} , the absolute change is

$$\Delta D = D_{t_2} - D_{t_1}. \quad (25)$$

For example, from March to June 2025,

$$\Delta D = 0.582294 - 0.444966 \approx 0.137328 \quad (26)$$

percentage points.

This is different from a relative percentage change. The relative change would be

$$\frac{D_{t_2} - D_{t_1}}{D_{t_1}} \times 100. \quad (27)$$

Keeping these two concepts separate avoids a common error in financial reporting.

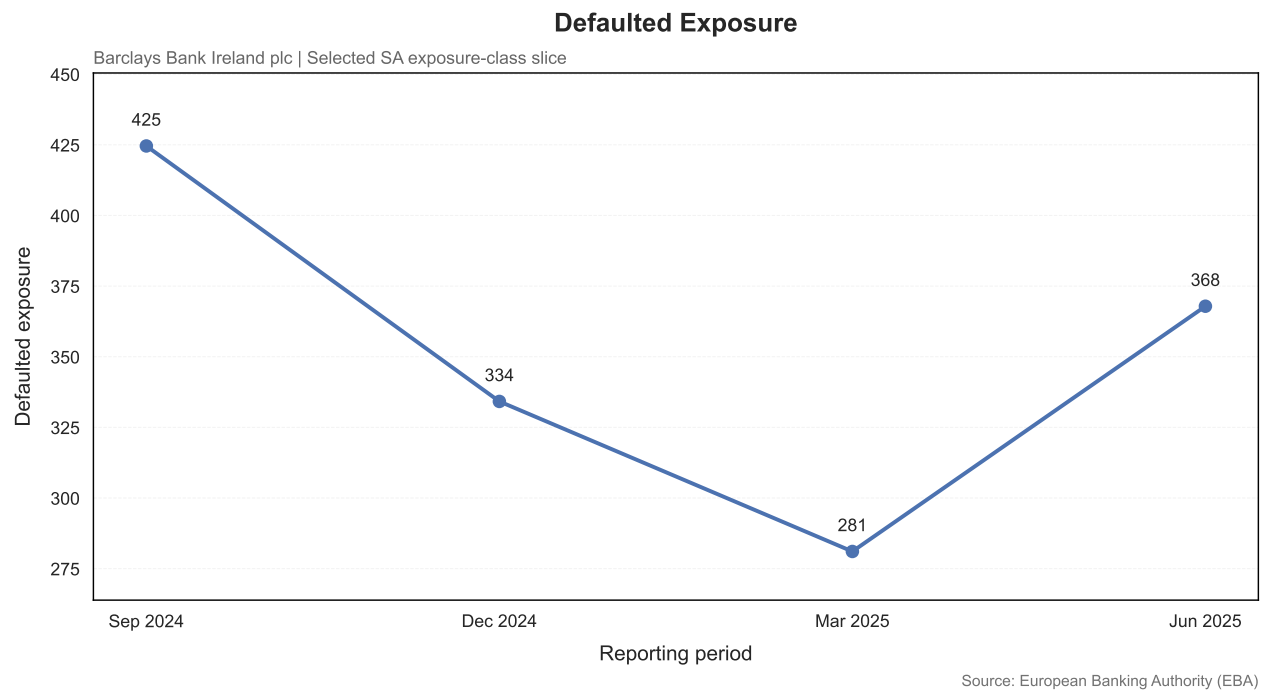


FIG. 3. **Absolute defaulted exposure across the four EBA reporting periods used in the project.** The amount falls from approximately 425 in September 2024 to 281 in March 2025, before rising to approximately 368 in June 2025. This graph displays the numerator of the defaulted-exposure-share calculation. By itself it does not account for changes in the size of the overall selected exposure base, which is why the relative measure in Fig. 4 is also examined.

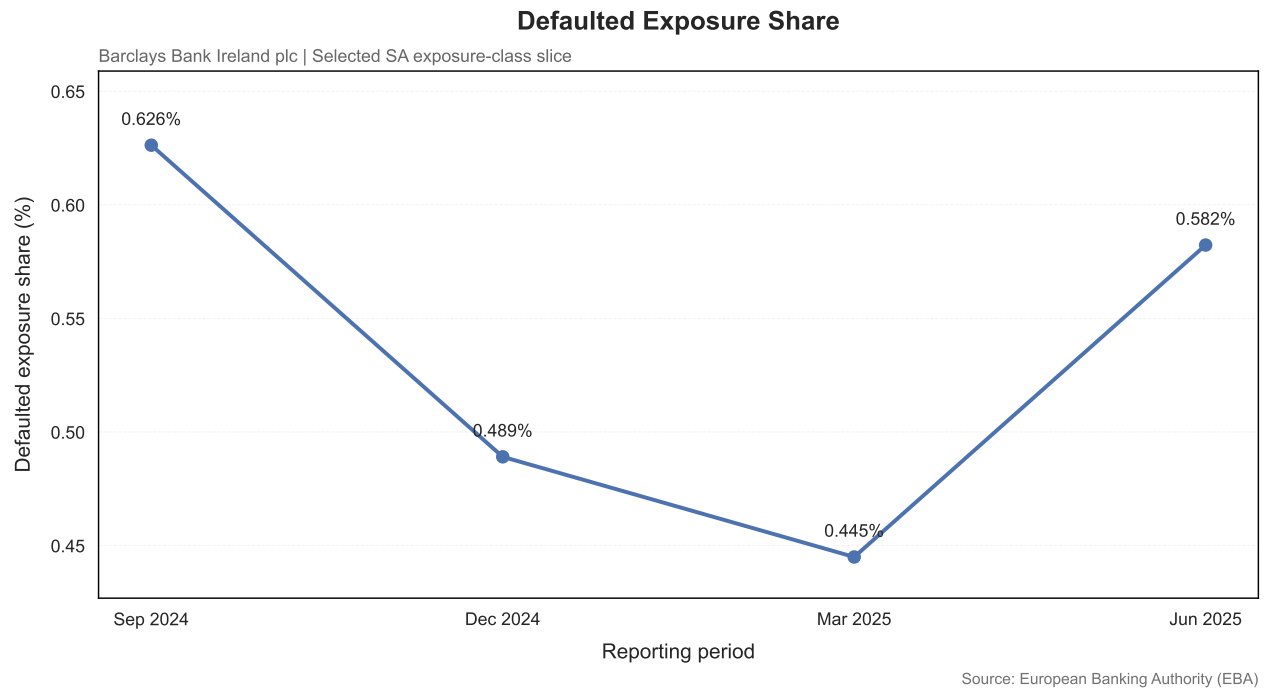


FIG. 4. **Defaulted exposure as a percentage of selected Standardised Approach exposure.** The ratio decreases from approximately 0.626% in September 2024 to 0.445% in March 2025 and then rises to approximately 0.582% in June 2025. Unlike Fig. 3, this graph normalizes the defaulted amount by the total selected exposure, making the values easier to compare across reporting dates. The ratio is descriptive and must not be interpreted as a probability of default.

VII. PEER COMPARISON

A. What is the purpose of a peer comparison?

A number for one bank is easier to interpret when it is placed beside numbers calculated using the same method for other institutions. The project therefore compares selected Irish and Dutch reporting institutions.

Each institution j is represented by two coordinates,

$$\mathbf{r}_j = (HHI_j, D_j). \quad (28)$$

This is the mathematical form of the peer scatter plot.

The horizontal coordinate, HHI_j , measures exposure-class concentration. The vertical coordinate, D_j , measures the defaulted-exposure share.

B. How to read the scatter plot

Moving to the right means that the selected exposure distribution is more concentrated. Moving upward means that the observed defaulted-exposure share is larger.

These axes measure different features. A point can therefore be far to the right but low on the vertical axis, meaning high concentration but a relatively low defaulted-exposure share. Conversely, an institution can have a less concentrated exposure distribution but a higher defaulted-exposure share.

The plot is therefore not a league table in which the upper-right or lower-left position automatically determines whether one bank is “good” or “bad.” Differences in business model, portfolio structure, geography, and regulatory reporting composition can affect both variables.

C. Peer median

The project adds median reference lines for the Irish peer group. The median is the middle observation after values are sorted.

For odd N ,

$$\tilde{x} = x_{(\frac{N+1}{2})}. \quad (29)$$

For even N ,

$$\tilde{x} = \frac{x_{(\frac{N}{2})} + x_{(\frac{N}{2}+1)}}{2}. \quad (30)$$

For a simple example, the median of

$$1, 2, 3, 8, 20 \quad (31)$$

is 3. The very large value 20 does not pull the median upward in the way it would affect an arithmetic mean.

This robustness is useful in a small peer sample where one institution may have an unusually high or low observation.

The resulting two-dimensional comparison is shown in Fig. 5. Each point represents one reporting institution. The horizontal coordinate is HHI and the vertical coordinate is the defaulted-exposure share. Barclays Bank Ireland is highlighted separately so that its position can be read against the Irish and Dutch reporting groups. The dashed reference lines are peer medians used only as descriptive guides; they are not regulatory limits or risk thresholds.

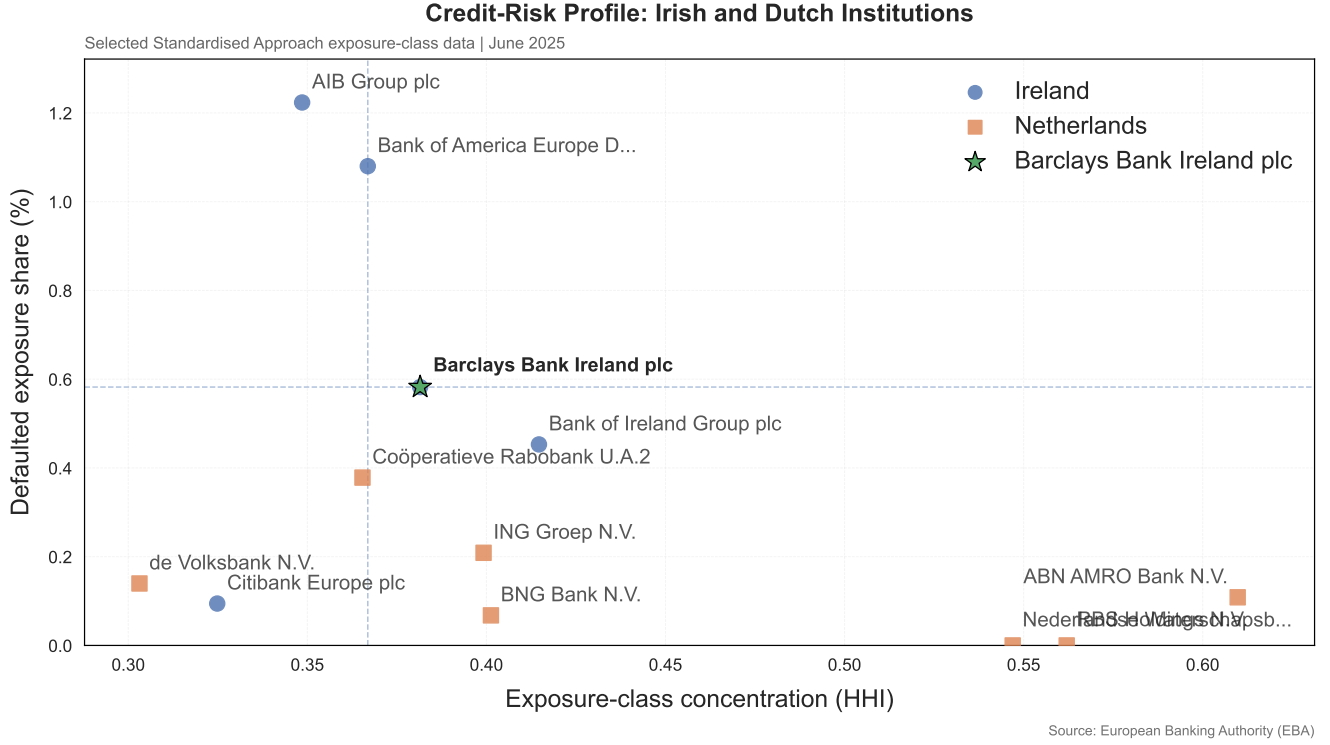


FIG. 5. **Peer comparison of exposure-class concentration and defaulted-exposure share for selected Irish and Dutch reporting institutions in June 2025.** The horizontal axis is the HHI of the selected SA exposure-class distribution, so points farther to the right have more concentrated exposure-class structures. The vertical axis is the defaulted-exposure share, so points higher on the graph have a larger reported defaulted amount relative to their selected exposure base. Barclays Bank Ireland is marked separately. The dashed lines show Irish peer medians and serve only as visual reference points. The chart is not a league table: differences in business model, portfolio mix, geography, and regulatory reporting structure can influence the location of each institution.

VIII. HYPOTHETICAL STRESS SENSITIVITY

A. What is being stressed?

The final part of the project asks a deliberately simple “what-if” question:

If the amount of defaulted exposure were larger than the June 2025 baseline, what would happen to the defaulted-exposure share if the selected total exposure were kept fixed?

This is a sensitivity exercise. It is designed to show how an output changes when one input assumption is changed. It is not an attempt to predict which stress scenario will actually occur.

B. Stress multiplier

Let

$$m \geq 1 \tag{32}$$

be a multiplier applied to baseline defaulted exposure.
The stressed defaulted exposure is

$$E_{\text{def}}(m) = mE_{\text{def},0}. \tag{33}$$

If $m = 1$, nothing changes. If $m = 1.25$, defaulted exposure is assumed to be 25% higher than baseline. If $m = 2$, it is assumed to double.

For example,

$$1.25 \times 367.874881 = 459.843602. \quad (34)$$

C. Why total exposure is held constant

The model assumes

$$E_{\text{tot}}(m) = E_{\text{tot},0}. \quad (35)$$

Conceptually, this can be read as a simplified reclassification or migration exercise: more of the already selected exposure is treated as defaulted, rather than new exposure being added to the portfolio.

This assumption is intentionally simple. A real stress-testing framework could allow total exposure, portfolio composition, collateral values, macroeconomic conditions, default probabilities, recoveries, and many other quantities to change simultaneously.

D. Stressed defaulted-exposure share

Under this assumption,

$$D(m) = \frac{mE_{\text{def},0}}{E_{\text{tot},0}} \times 100. \quad (36)$$

The baseline share is

$$D_0 = \frac{E_{\text{def},0}}{E_{\text{tot},0}} \times 100. \quad (37)$$

Therefore,

$$\boxed{D(m) = mD_0}. \quad (38)$$

This equation has a simple interpretation: because the denominator is fixed, multiplying defaulted exposure by 1.5 also multiplies the defaulted-exposure share by 1.5.

E. Discrete scenarios

The project uses the following illustrative assumptions:

Scenario	Multiplier m	Defaulted exposure	Share (%)
Baseline	1.00	367.87	0.582
Mild stress	1.25	459.84	0.728
Moderate stress	1.50	551.81	0.873
Severe stress	2.00	735.75	1.165

The labels “mild,” “moderate,” and “severe” are convenient names for the hypothetical multipliers used in this educational exercise. They are not EBA-defined scenarios and are not calibrated from macroeconomic forecasts.

IX. SENSITIVITY COEFFICIENT

A. What does sensitivity mean?

Sensitivity asks how quickly an output changes when an input changes. Here the input is the stress multiplier m , and the output is the defaulted-exposure share D .

The mathematical sensitivity is the derivative

$$\frac{\partial D}{\partial m}. \quad (39)$$

Since

$$D(m) = mD_0, \quad (40)$$

the derivative is

$$\boxed{\frac{\partial D}{\partial m} = D_0}. \quad (41)$$

For June 2025,

$$\frac{\partial D}{\partial m} \approx 0.582294 \quad (42)$$

percentage points per unit change in the multiplier.

B. Interpreting the derivative without calculus

The derivative may be understood simply as the slope of the sensitivity line. Because the model is linear, the slope is constant.

Suppose the multiplier increases from 1.00 to 1.25:

$$\Delta m = 0.25. \quad (43)$$

The corresponding change in the defaulted-exposure share is

$$\Delta D = D_0 \Delta m \quad (44)$$

$$\approx 0.582294 \times 0.25 \quad (45)$$

$$\approx 0.145574 \quad (46)$$

percentage points.

Adding this to the baseline,

$$0.582294 + 0.145574 \approx 0.727868\%. \quad (47)$$

This is the mild-stress result.

X. CONTINUOUS SENSITIVITY SCAN

The discrete scenarios show only four selected points. To see the complete relationship, the project also evaluates many multiplier values over

$$m \in [1, 3]. \quad (48)$$

For every multiplier, the program calculates

$$D(m) = mD_0. \quad (49)$$

Plotting these values produces a straight line. The discrete baseline, mild, moderate, and severe scenarios lie on the same line.

The continuous curve is useful because it shows that the results are not generated by four unrelated assumptions. They are points from one mathematical response function. Figure 6 displays this relationship. The named baseline, mild, moderate, and severe scenarios lie on the same straight sensitivity curve because the selected total exposure is held fixed and only the defaulted-exposure amount is multiplied.

The straight line also reveals an important limitation: the model assumes that the relationship remains linear. Real credit losses and defaults can respond nonlinearly to economic stress, so this graph should be interpreted only as a controlled sensitivity experiment.

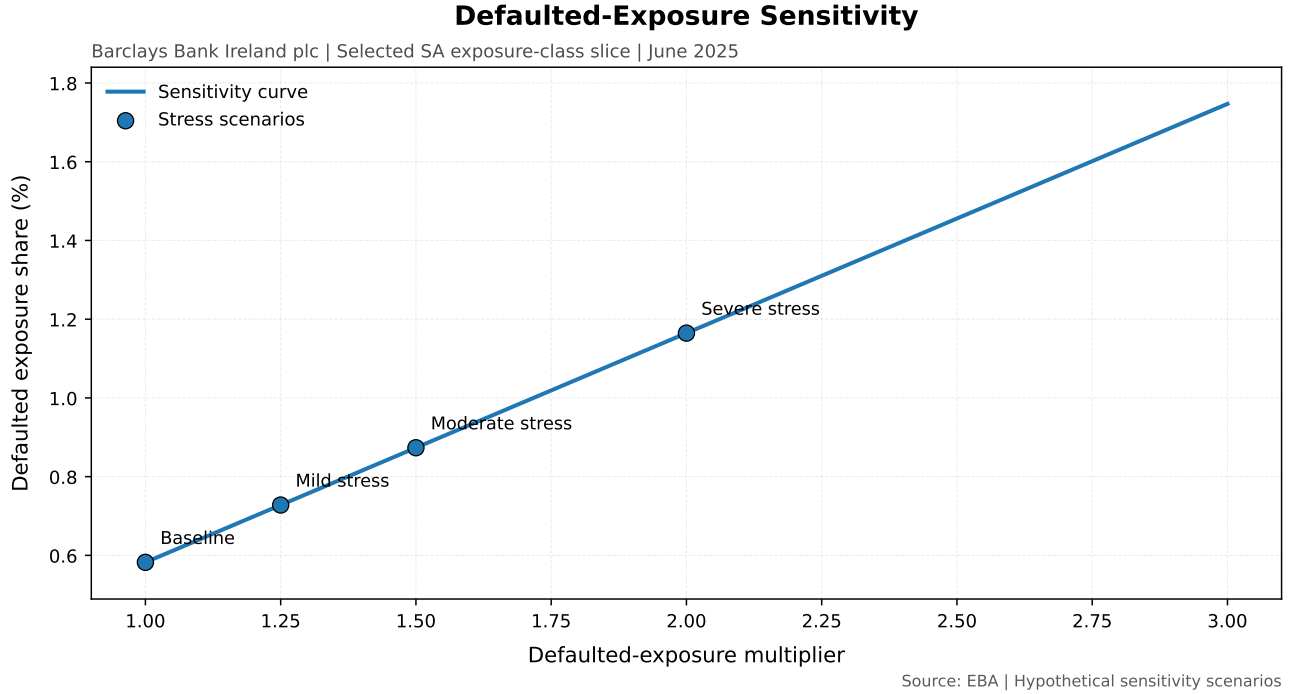


FIG. 6. **Hypothetical sensitivity of the defaulted-exposure share to increases in defaulted exposure.** The horizontal axis is a multiplier applied to the June 2025 baseline defaulted exposure, while the vertical axis is the resulting defaulted-exposure share. The baseline multiplier is 1.00; 1.25 represents a 25% increase, 1.50 a 50% increase, and 2.00 a doubling of the defaulted amount. The named stress scenarios are illustrative points on the continuous response curve. Because total selected exposure is held constant, the relationship is linear by construction. The figure is a mathematical sensitivity study, not an economic forecast or an official EBA stress scenario.

XI. CONNECTION WITH STANDARD CREDIT-RISK TERMINOLOGY

Readers entering finance will frequently encounter three quantities: probability of default, loss given default, and exposure at default.

A. Probability of default

Probability of default, or PD , estimates the probability that a borrower will default over a specified horizon.

B. Loss given default

Loss given default, or LGD , represents the proportion of exposure expected to be lost if default occurs, after considering recoveries.

C. Exposure at default

Exposure at default, or EAD , estimates the amount that would be exposed at the time of default. A simplified expected-loss relationship is often written as

$$EL = PD \times LGD \times EAD. \quad (50)$$

This equation is included to place the project in financial context. The public aggregate EBA dataset used here does not provide the borrower-level information required to estimate these parameters in the manner of an internal bank credit-risk model.

Therefore, this project does *not* calculate PD , LGD , EAD , or expected loss. Its defaulted-exposure share should not be substituted for PD in the equation above.

XII. WHAT IS NOT MODELLED

The analysis deliberately stops at transparent descriptive and sensitivity measures. It does not model

- borrower-level creditworthiness;
- credit scores or internal ratings;
- transition probabilities between rating states;
- probability of default;
- loss given default;
- exposure at default;
- collateral or recovery processes;
- expected credit loss;
- regulatory capital;
- correlations between individual borrowers;
- macroeconomic scenario paths; or
- future bank performance.

This limitation is important rather than problematic. The purpose of the project is to demonstrate a clean quantitative workflow using the information actually available in the public dataset.

XIII. HOW THE MATHEMATICS MAPS TO THE PYTHON PROJECT

The mathematical calculations correspond directly to the project scripts:

Script and Main quantitative task

`01_data_exploration.py`

Inspect the dataset, identify reporting periods and variables, and extract the selected bank.

`02_exposure_analysis.py`

Calculate and visualize exposure values and exposure-class composition.

`03_risk_trend.py`

Calculate total exposure, defaulted exposure, and the defaulted-exposure share through time.

`04_concentration_analysis.py`

Calculate HHI, top-three concentration, and largest-class share.

`05_peer_comparison.py`

Calculate the same descriptive indicators for selected peer institutions and compare them in HHI–default-share space.

`06_stress_analysis.py`

Apply hypothetical defaulted-exposure multipliers and calculate the resulting sensitivity curve.

This mapping is useful for readers who want to move between the equations in this document and their implementation in Python.

XIV. SUMMARY

The complete analytical logic can be summarized as

public EBA data → exposure composition → default ratio → concentration → time comparison → peer comparison → sensitivity (51)

The principal equations are

$$s_i = \frac{E_i}{E_{\text{tot}}}, \quad (52)$$

$$D = 100 \frac{E_{\text{def}}}{E_{\text{tot}}}, \quad (53)$$

$$HHI = \sum_i s_i^2, \quad (54)$$

$$C_k = 100 \sum_{j=1}^k s_{(j)}, \quad (55)$$

$$D(m) = mD_0, \quad (56)$$

$$\frac{\partial D}{\partial m} = D_0. \quad (57)$$

For a novice reader, the central idea is that each equation answers a different question. Exposure shares describe *where the portfolio is*. The defaulted-exposure ratio describes *how much of the selected exposure is reported as defaulted*. HHI and top- k ratios describe *how concentrated the exposure structure is*. Time-series calculations describe *how these quantities change*. Peer analysis supplies *context*. Finally, the stress calculation asks *how one selected indicator responds to a controlled hypothetical change*.

Together, these methods form a transparent introductory quantitative study of public bank credit-risk data while keeping a clear boundary between descriptive analysis and the more sophisticated credit-risk models used inside financial institutions.